Program: **Diploma in Artificial Intelligence & Machine Learning**	
Course Code: **5349B**	Course Title: **Fundamentals of Internet of Things Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Provide skills to build Internet of Things (IoT) systems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic electronics	2039	Fundamentals of Electrical and Electronics Engineering Lab	II
Programming Concepts	2139	Problem Solving and Programming Lab	II
Programming Concepts	3347	Python Programming Lab	III

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Write Python programs for IoT applications	12	Apply
CO2	Develop simple IoT applications using computing boards (Node MCU/Arudino) and sensors	12	Apply
CO3	Develop IoT applications using Raspberry Pi	13	Apply
CO4	Design use cases and develop IoT applications for various domains	5	Create
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3						
CO3	3						
CO4	3	3	3	3	3	3	3

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	**Write Python programs for IoT applications**		
M1.01	Write Python programs that demonstrate the use of various data types, including lists, tuples, and dictionaries.	4	Apply
M1.02	Write programs using control structures in Python	4	Apply
M1.03	Develop Programs using functions, packages and modules	4	Apply
CO2	**Develop simple IoT applications using computing boards (Node MCU/Arudino/ESP32 etc) and sensors**		
M2.01	Create a tinker CAD program to implement a Traffic Light Controller using 3 different LEDs	3	Apply
M2.02	Create a Tinker cad program to implement an LED Light show Create a Tinker CAD program to read room temperature using a temperature sensor and print the temperature using an LED display Create a program to read room temperature and switch on a servo motor if greater than a threshold value	6	Apply
M2.03	Develop applications to control sensors through web pages and mobile applications	3	Apply
	Lab Exam – I	1.5	

CO3	**Develop IoT applications using Raspberry Pi**		
M3.01	Develop an IoT application to continuously monitor and measure the moisture level of soil using a moisture sensor and switch on the sprinkler	4	Apply
M3.02	Develop IoT applications with Raspberry PI to interact with web, mobile applications and cloud (RFID-based toll collection system)	8	Apply
CO4 : Open Ended experiments to Design use cases and develop an IoT application for various domains			
M4.01	Implement an IoT-based health care system which continuously measures the air quality and sends reports to the asthma patients about the environment, also measures the breathing variations and provides alarms to use inhalers. Implement an IoT-based car parking system	6	Apply
	Lab Exam – II	1.5	

Text / Reference

T/R	Book Title/Author
R1	Gary Smart, **Practical Python Programming for IoT,** Packt Publishing
R2	Marco Schwartz, **Internet of Things with Arduino Cookbook,** Packt Publishing

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/106/105/106105166/
2	https://www.raspberrypi.org/blog/getting-started-with-iot/
3	https://learn.adafruit.com/category/internet-of-things-iot
4	https://www.arduino.cc/en/IoT/HomePage
5	http://esp32.net/